Blood Groups

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Blood is a vital body fluid containing plasma, platelets, red blood cells and white blood cells. In 1901, Karl Landsteiner formulated a blood classification system and also discovered that blood groups were inherited in a way that agreed with Mendel's laws of inheritance. In the ABO system, blood is classified depending upon whether the antigens A or B are present in red blood cells. Antigens are complex chemical substances.



Antigens: substances that cause the production of antibodies (defence proteins) to combat infection.

Group A blood has the A antigen, group B has the B antigen, group AB has both and group O has neither. This is because the gene for blood type has three alleles: I^A, I^B and I^O.

1 What is an 'allele'? _____

The genes I^A and I^B are dominant over I^O. However gene IA is only incompletely dominant over I^B.

Genes	Blood type
$I^A + I^A$	A
$I^A + I^O$	A
$I^O + I^O$	O
$I^B + I^B$	B
$I^A + I^B$	AB
$I^B + I^O$	B

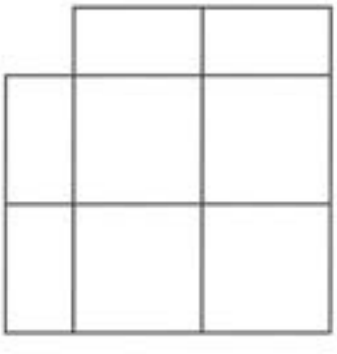
Because there are only two chromosomes of each type per cell, there can be only two alleles present at any one time.

The diagram on the right shows the blood groups that can be formed.

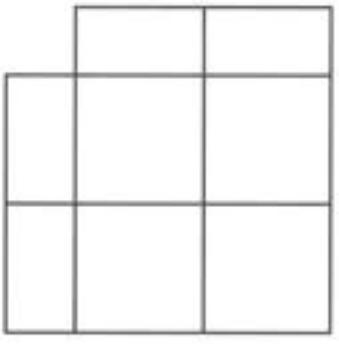
2 How many different blood groups are there according to this system?

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3 Use the Punnett square on the right to show the possible blood groups of offspring, inherited from an I^AI^B mother and an I^BI^O father.



4 Use the Punnett square on the right to show the possible blood groups of offspring, inherited from an I^AI^A mother and an I^AI^B father.



5 Who was the person who first classified blood types as A, B, AB and O?

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6 Suggest two examples of when blood types might be used in legal cases.

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